Held-out MCC-P across all datasets

Gemma-2-2B

Dataset	SSAE L1	SSAE Step-L0	SSAE sigmoid-L0	SSAE ReLU	SSAE JumpReLU	SSAE TopK	SAE ReLU	SAE JumpReLU	SAE TopK
ENG→FR	0.918 ± 0.007	0.969 ± 0.001	0.973 ± 0.000	0.170 ± 0.106	0.887 ± 0.019	0.822 ± 0.023	0.275 ± 0.161	0.850 ± 0.055	0.905 ± 0.013
ENG→DE	0.891 ± 0.027	0.970 ± 0.000	0.971 ± 0.001	0.281 ± 0.029	0.918 ± 0.007	0.913 ± 0.008	0.074 ± 0.048	0.908 ± 0.026	0.910 ± 0.029
GENDER	0.991 ± 0.005	0.990 ± 0.002	0.994 ± 0.002	0.126 ± 0.135	0.902 ± 0.010	0.854 ± 0.032	0.314 ± 0.206	0.871 ± 0.025	0.881 ± 0.032
BINARY	0.799 ± 0.008	0.753 ± 0.008	0.804 ± 0.010	0.090 ± 0.018	0.753 ± 0.027	0.637 ± 0.117	0.119 ± 0.056	0.693 ± 0.054	0.666 ± 0.036
CORR	0.882 ± 0.016	0.750 ± 0.030	0.885 ± 0.022	0.148 ± 0.088	0.846 ± 0.015	0.853 ± 0.012	0.265 ± 0.063	0.827 ± 0.013	0.799 ± 0.010
TruthfulQA	0.452 ± 0.014	0.602 ± 0.016	0.598 ± 0.036	0.074 ± 0.052	0.379 ± 0.067	0.340 ± 0.057	0.170 ± 0.002	0.470 ± 0.047	0.501 ± 0.078
Sycophancy	0.629 ± 0.005	0.190 ± 0.093	0.633 ± 0.031	0.072 ± 0.005	0.151 ± 0.070	0.235 ± 0.104	0.108 ± 0.116	0.198 ± 0.138	0.135 ± 0.056
Refusal	0.802 ± 0.074	0.843 ± 0.038	0.782 ± 0.053	0.069 ± 0.066	0.660 ± 0.142	0.382 ± 0.073	0.469 ± 0.172	0.682 ± 0.135	0.388 ± 0.251
Bias-in-Bios	0.334 ± 0.004	0.241 ± 0.001	0.216 ± 0.002	0.429 ± 0.005	0.154 ± 0.004	0.502 ± 0.003	0.464 ± 0.005	0.257 ± 0.005	0.527 ± 0.012

Pythia-70M

Dataset	SSAE L1	SSAE Step-L0	SSAE sigmoid-L0	SSAE ReLU	SSAE JumpReLU	SSAE TopK	SAE ReLU	SAE JumpReLU	SAE TopK
ENG→FR	0.744 ± 0.039	0.844 ± 0.030	0.913 ± 0.015	0.238 ± 0.146	0.800 ± 0.009	0.488 ± 0.157	0.103 ± 0.048	0.540 ± 0.085	0.735 ± 0.028
ENG→DE	0.750 ± 0.092	0.782 ± 0.035	0.945 ± 0.017	0.491 ± 0.056	0.830 ± 0.019	0.508 ± 0.085	0.239 ± 0.075	0.594 ± 0.117	0.693 ± 0.071
GENDER	0.550 ± 0.222	0.808 ± 0.081	0.842 ± 0.038	0.205 ± 0.054	0.709 ± 0.063	0.291 ± 0.180	0.410 ± 0.060	0.570 ± 0.154	0.415 ± 0.091
BINARY	0.421 ± 0.101	0.592 ± 0.035	0.606 ± 0.073	0.104 ± 0.016	0.546 ± 0.008	0.125 ± 0.077	0.170 ± 0.013	0.459 ± 0.076	0.274 ± 0.099
CORR	0.623 ± 0.069	0.562 ± 0.033	0.682 ± 0.038	0.139 ± 0.090	0.596 ± 0.021	0.328 ± 0.037	0.151 ± 0.040	0.497 ± 0.081	0.404 ± 0.076
TruthfulQA	0.178 ± 0.055	0.139 ± 0.063	0.529 ± 0.038	0.204 ± 0.017	0.340 ± 0.059	0.211 ± 0.045	0.247 ± 0.006	0.397 ± 0.014	0.355 ± 0.061
Sycophancy	0.456 ± 0.074	0.384 ± 0.001	0.628 ± 0.068	0.385 ± 0.000	0.466 ± 0.068	0.401 ± 0.017	0.386 ± 0.000	0.729 ± 0.019	0.376 ± 0.010
Refusal	0.355 ± 0.027	0.083 ± 0.055	0.517 ± 0.041	0.154 ± 0.020	0.519 ± 0.155	0.309 ± 0.103	0.161 ± 0.132	0.244 ± 0.031	0.295 ± 0.080
Bias-in-Bios	0.172 ± 0.003	0.124 ± 0.002	0.173 ± 0.002	0.206 ± 0.007	0.140 ± 0.003	0.194 ± 0.004	0.265 ± 0.004	0.143 ± 0.004	0.243 ± 0.004